

TECHNICAL REQUIREMENTS DOCUMENT

Professional AC Core Cutting Business Website

TRD v1.0 • Production Technical Specification

System	AC Core Cutting Business Website
Architecture	Modern server-rendered / hybrid web application
Recommended Stack	Next.js + TypeScript + Tailwind CSS
Primary Goal	Fast, secure, SEO-ready local lead generation
Deployment	Production cloud hosting + custom domain + HTTPS
Status	Technical baseline finalized

1. Technical Objectives

- Deliver a fast, responsive and maintainable production website.
- Use server rendering/static generation where it improves SEO and performance.
- Keep client-side JavaScript minimal and intentional.
- Provide reliable Call, WhatsApp and enquiry conversion paths.
- Protect enquiry forms and minimize collection of unnecessary personal data.
- Make services, locations, reviews and gallery content easy to extend.

2. Recommended Technology Stack

Layer	Technology / Standard	Purpose
Framework	Next.js + TypeScript	Application framework, routing, rendering and SEO foundation.
UI	React components	Reusable interface components.
Styling	Tailwind CSS	Responsive design system and utility styling.
Validation	Zod or equivalent	Typed input validation for forms/API boundaries.
Icons	Lucide or equivalent	Consistent lightweight iconography.
Images	Next/Image or equivalent	Responsive image optimization.
Forms	Server/API endpoint	Secure lead submission and validation.
Hosting	Production cloud platform	HTTPS, CDN and deployment pipeline.
Analytics	Privacy-conscious analytics	Measure conversions and traffic.

3. Application Architecture

Browser

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■■■ Responsive UI / accessible components

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Next.js Application

■■■ Public pages
■■■ Service/location routes
■■■ SEO metadata + structured data
■■■ Server-rendered/static content
■■■ API / server action for enquiries
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■■■ Validation + spam protection

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Lead destination

(email / approved CRM / business workflow)

4. Routing & URL Structure

Route	Purpose
/	Home
/services	Services overview
/services/ac-core-cutting	AC Core Cutting

Route	Purpose
/services/rcc-core-cutting	RCC Core Cutting
/services/ac-drain-hole	AC Drain Hole
/services/concrete-wall-drilling	Concrete Wall Drilling
/services/pipe-cable-passage	Pipe & Cable Passage
/services/other	Other Services
/about	About
/gallery	Gallery
/reviews	Reviews
/faq	FAQ
/service-areas	Service Areas
/contact	Contact

5. Component Architecture

- **Layout:** Header, mobile navigation, footer and page container.
- **Navigation:** Desktop navigation + accessible mobile menu.
- **Hero:** Service headline, supporting copy and primary CTAs.
- **CTA:** Reusable Call, WhatsApp and Quote buttons.
- **ServiceCard:** Consistent service preview with route and CTA.
- **TrustSection:** Experience, equipment, accuracy and cleanliness claims based on verified business information.
- **GalleryGrid:** Responsive image grid with optimized media.
- **ReviewCard:** Approved customer testimonial display.
- **FAQAccordion:** Keyboard-accessible expandable questions.
- **ContactForm:** Validated lead form with loading, success and error states.
- **ServiceAreaList:** Structured list of genuinely served locations.
- **MapBlock:** Map/location embed or external map link.

6. Content Model

Entity	Key Fields
Business	name, logo, phone, whatsapp, address, hours, city, serviceAreas, description
Service	slug, name, summary, description, benefits, process, image, seoTitle, seoDescription
GalleryItem	image, altText, title, category, location, date(optional)
Review	customerName, reviewText, rating(optional), source, date(optional), approved
FAQ	question, answer, category, order
ServiceArea	name, slug, description(optional), mapReference(optional)

7. Form / Lead API Specification

- **Method:** POST over HTTPS.
- **Suggested fields:** name, phone, WhatsApp preference, service type, location, message.
- **Validation:** Required fields, phone format, maximum lengths and safe character handling.
- **Spam protection:** Rate limiting and/or honeypot/CAPTCHA-equivalent where necessary.
- **Response:** Return a clear success or safe generic error; do not expose internal errors.
- **Logging:** Log operational failures without storing unnecessary sensitive information.
- **Destination:** Email, approved CRM or business workflow chosen during implementation.

8. SEO & Rendering Architecture

- Use server-rendered/static content for public service pages whenever possible.
- Generate unique metadata for every indexable page.
- Use canonical URLs and clean, descriptive slugs.
- Generate XML sitemap and robots.txt.
- Add relevant LocalBusiness and Service structured data using verified information.

- Use semantic HTML headings and internal links.
- Provide descriptive image alt text; avoid keyword stuffing.
- Add Open Graph/Twitter-style social metadata.
- Prevent accidental indexing of internal/error-only routes.

9. Performance Requirements

Area	Requirement
Images	Use modern responsive formats where supported; size images to actual display needs.
JavaScript	Minimize client components and third-party scripts.
Fonts	Limit font families/weights and use efficient loading.
Caching	Cache static/public assets appropriately.
Network	Serve assets through CDN where available.
Core Web Vitals	Target strong LCP, INP and CLS results on representative mobile devices.
Third parties	Load maps/analytics/widgets only when needed and avoid unnecessary blocking.

10. Security Requirements

- Force HTTPS in production.
- Validate and sanitize all untrusted form input server-side.
- Do not expose secrets, API keys or private configuration in client-side code.
- Apply rate limiting/spam controls to enquiry endpoints.
- Use secure headers appropriate to the hosting architecture.
- Avoid storing unnecessary personal data.
- Restrict administrative credentials and services if a CMS/admin system is added.
- Keep dependencies updated and review production build warnings.

11. Accessibility Requirements

- Semantic HTML and logical heading hierarchy.
- Keyboard-accessible navigation, accordions and forms.
- Visible focus states.
- Accessible form labels and error messages.
- Meaningful alt text for informative images; decorative images marked appropriately.
- Sufficient text/background contrast.
- Buttons and links must have clear names and adequate touch target size.

12. Analytics & Conversion Tracking

- Track page views and key conversion events.
- Recommended events: phone_click, whatsapp_click, quote_start, quote_submit, map_click and service_cta_click.
- Do not collect unnecessary sensitive data through analytics.
- Document event names so future marketing/reporting remains consistent.

13. Deployment & Environments

Environment	Purpose
Local	Developer implementation and testing.

Environment	Purpose
Preview/Staging	Review content, UI, forms and QA before production.
Production	Public website on final domain with HTTPS and monitoring.

14. QA & Testing Plan

- **Functional:** Test navigation, links, CTAs, forms, validation, success/error states and maps.
- **Responsive:** Test common phone, tablet and desktop widths.
- **Browser:** Test current Chrome, Edge, Safari and Firefox where practical.
- **Accessibility:** Keyboard navigation, focus order, labels, contrast and screen-reader-friendly semantics.
- **SEO:** Verify titles, descriptions, canonicals, sitemap, robots, structured data and indexability.
- **Performance:** Run Lighthouse/PageSpeed-style checks and optimize major bottlenecks.
- **Security:** Test input validation, spam controls, secret exposure and HTTPS.
- **Content:** Verify phone/WhatsApp, service areas, prices (if shown), hours, reviews and business claims.

15. Repository / Project Structure

```
src/  
■■■ app/  
■ ■■■ page.tsx  
■ ■■■ services/  
■ ■■■ about/  
■ ■■■ gallery/  
■ ■■■ reviews/  
■ ■■■ faq/  
■ ■■■ service-areas/  
■ ■■■ contact/  
■ ■■■ api/  
■■■ components/  
■■■ content/  
■■■ lib/  
■■■ styles/  
■■■ types/  
  
public/  
■■■ images/  
■■■ icons/  
■■■ ...
```

16. Technical Acceptance Criteria

- Production build completes without critical errors or warnings that affect functionality.
- All approved routes return correctly and have intended metadata.
- No broken internal links or missing required assets.
- Forms work over HTTPS and handle invalid, successful and failed submissions safely.
- No private credentials/secrets are shipped to the browser.
- Responsive layout works across agreed device sizes.
- Accessibility baseline is met for navigation and forms.
- Performance has been reviewed and major bottlenecks addressed.
- Sitemap, robots and structured data are present and valid.

■ Production domain, HTTPS and monitoring/analytics are configured.

17. Implementation Notes

- Business-specific phone, WhatsApp, address, service areas, hours and claims must come from the business owner.
- Real project photographs should be optimized before upload and given meaningful alt text.
- Pricing should not be published unless the business has stable pricing rules and wants public pricing.
- Maps and external review widgets should be added only when they provide clear value and do not materially hurt performance.
- If a CMS is introduced later, the content model in this TRD can become the basis for the CMS schema.

TRD conclusion: The technical architecture prioritizes performance, local SEO, maintainability, accessibility, security and conversion. The stack can support a professional MVP now and future expansion later.